

**Childcare Accessibility in San Francisco:
A Two-Step Floating Catchment Area Analysis of Early Childhood
Facility Access**

SSCI 583: Spatial Analysis | Project 3

USC Spatial Sciences Institute

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1. Introduction

Spatial accessibility refers to the ability of individuals to reach services based on their location, the availability of those services, and the effort required to travel to them. In urban environments, accessibility is shaped not only by the distribution of facilities but also by the transportation network that connects people to those services. The two-step floating catchment area (2SFCA) method is commonly used to measure accessibility because it accounts for both supply (service locations) and demand (population), while also incorporating travel distance or time.

This project evaluates accessibility to early childhood education facilities in San Francisco, focusing on children ages 0–5. The goal is to understand how well daycare and preschool facilities are spatially distributed relative to the population that needs them. The analysis applies the 2SFCA method using a network-based approach to identify areas with high and low accessibility and to assess potential inequalities in childcare availability across the city.

2. Study Area

The study area is the City and County of San Francisco, a dense and geographically constrained urban environment located on a peninsula in Northern California. The city's compact form and highly developed street network make it suitable for network-based accessibility analysis. However, the uneven distribution of residential areas and services can create disparities in access to essential facilities such as childcare.

The study area map illustrates the distribution of children ages 0–5 across census block groups, along with the locations of early childhood education facilities. This map provides an initial understanding of where demand is concentrated and how it relates spatially to available facilities. The supply side of the analysis consists of licensed early childhood education facilities — including daycare centers and preschool programs — operating within San Francisco County. As shown in Figure 2 (Facility Map), these facilities are distributed across the city, with notable clusters in the northeastern neighborhoods, the Mission, and the Bayview. However, visually comparing the facility map with the child population density map suggests a partial mismatch: some of the areas with the highest concentrations of young children, particularly in Bayview-Hunters Point, appear to have relatively fewer facilities relative to their population size.

Early Childhood Educational Facilities in San Francisco

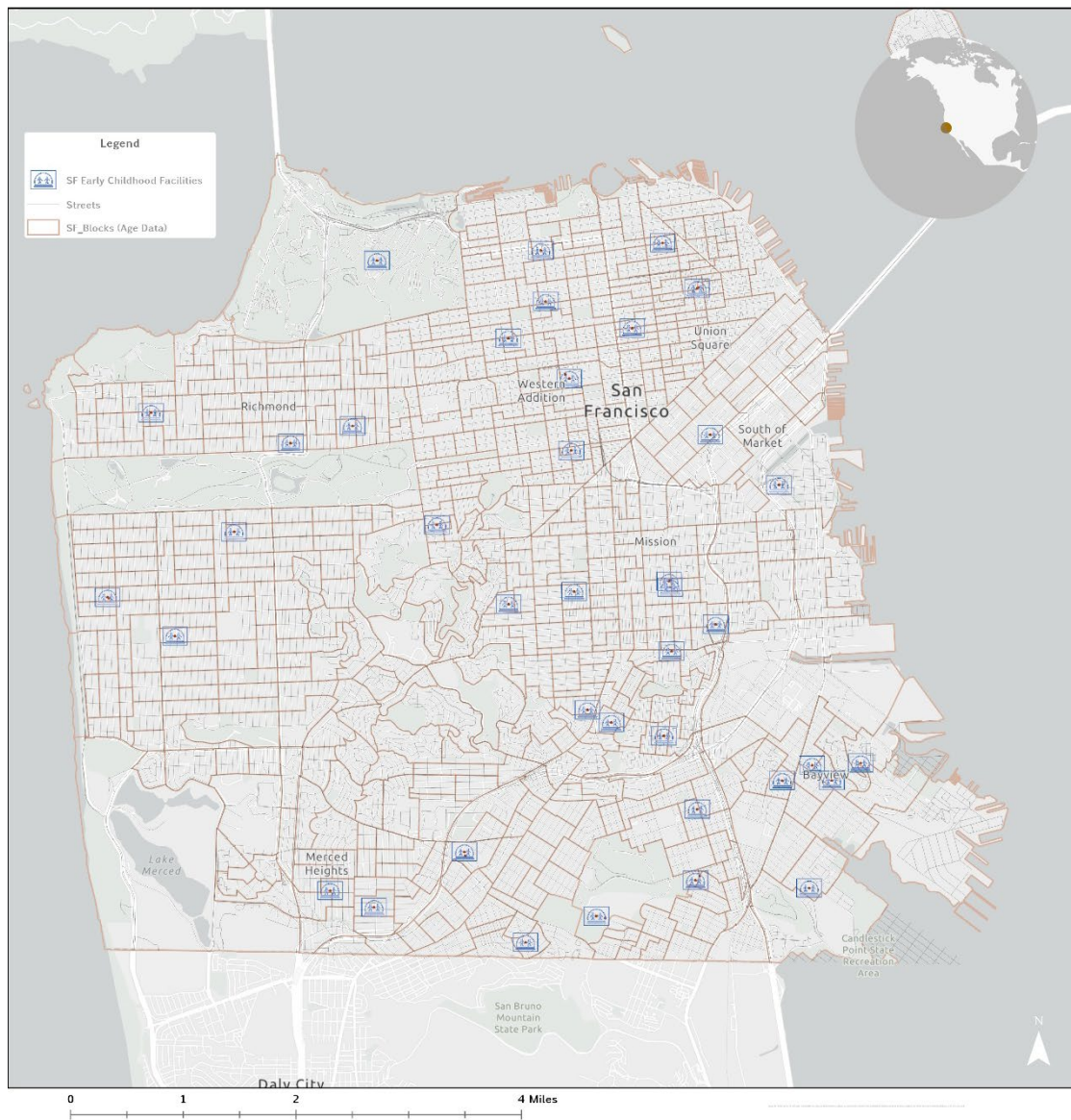


Figure 1. Population Density for Children Ages 0–5 by Census Block, San Francisco County. Darker shades indicate higher concentrations of children ages 0–5. Early childhood facility locations are shown as symbols. Source: U.S. Census Bureau, 2019 ACS; DataSF.

Facility map

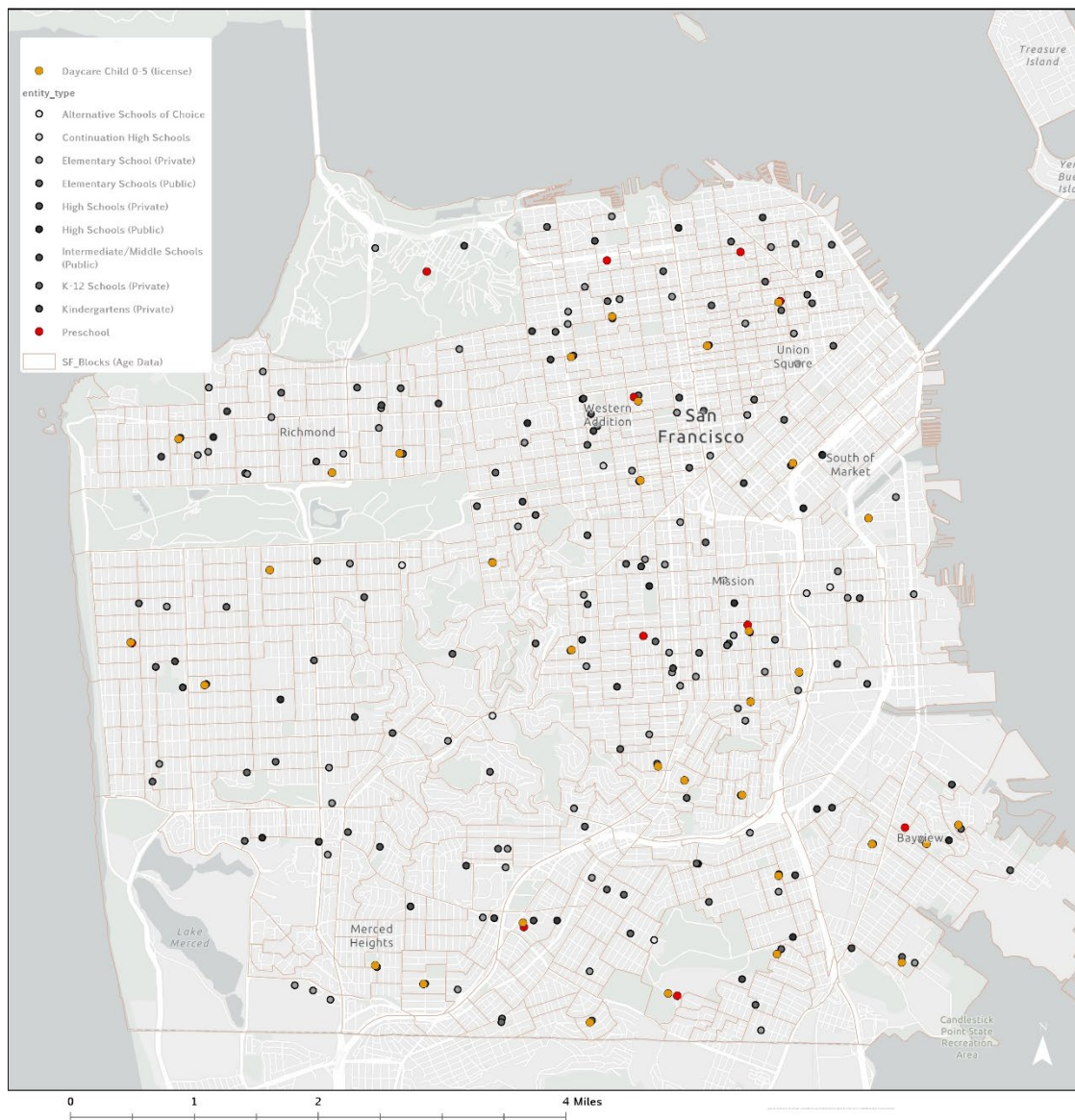


Figure 2. Facility Map — Early Childhood Educational Facilities in San Francisco. Orange circles indicate licensed daycare centers (ages 0–5); additional education facility types are shown for reference. Source: DataSF.

3. Methods

3.1 Data

The analysis draws on three primary data sources, summarized in Table 1. Population demand data were obtained from the U.S. Census Bureau's 2019 American Community Survey five-year estimates at the census block level. The specific variable used was the total count of

children ages zero to five (variable B01001), which was spatially joined to block polygons obtained from the Census TIGER/Line shapefiles. Block centroids were generated using ArcGIS Pro's Feature to Point tool and used as the population demand locations in the 2SFCA calculation.

Childcare facility data were downloaded from DataSF, San Francisco's open data portal. The dataset includes the locations of all licensed early childhood education facilities in the county, including daycare centers and preschool programs. Facilities were filtered to include only those licensed to serve children ages zero to five. Each facility was represented as a point feature. No capacity data were available in the source dataset; consequently, facility count (each facility weighted equally as one unit of supply, $S_j = 1$) was used as the supply variable, consistent with studies that lack detailed capacity records.

The street network used for travel time computation was drawn from the Esri Living Atlas Network Dataset for the San Francisco region. All layers were projected into NAD 1983 State Plane California III (2011) FIPS 0404 Feet to ensure spatial consistency.

Table 1. Data Summary

Data Layer	Source	Format	Description
SF Early Childhood Facilities	DataSF	Point feature class	Licensed daycare and preschool facilities serving children 0–5 in San Francisco County
SF Blocks – Population Age 0–5	US Census Bureau, 2019 ACS	Polygon / Point (centroids)	Total population of children ages 0–5 aggregated to census block level; centroids used as demand locations
San Francisco Street Network	Esri / ArcGIS Online Living Atlas	Network dataset	Road network used to compute 10-minute driving-time catchments via ArcGIS Pro Closest Facility solver

3.2 Workflow and Analytical Steps

The 2SFCA method proceeds in two sequential steps. In Step 1, a catchment is drawn around each facility location j , and all population locations k falling within that catchment are identified. The supply-to-demand ratio R_j is then calculated as:

$$R_j = S_j / \sum P_k \text{ (for all } k \text{ where } dkj \leq d_o)$$

where S_j is the supply at facility j (in this study, equal to 1 for each facility), P_k is the population of children ages 0–5 at block centroid k , and dkj is the travel time between block centroid k and facility j . Only population locations within the threshold travel time d_0 — here set to 10 minutes by driving — are included in the denominator.

In Step 2, a catchment of the same size is drawn around each population location i , and the R_j values of all facilities j within that catchment are summed to produce the final accessibility score:

$$A_i = \sum R_j \text{ (for all } j \text{ where } dij \leq d_0)$$

A larger A_i value indicates greater accessibility — the population location i is within reach of facilities that are not excessively burdened by competing demand. A small A_i value indicates that either few facilities are reachable, or those that are reachable serve very large competing populations.

Because the OD Cost Matrix tool was unavailable, the Closest Facility network solver in ArcGIS Pro was used as a methodological substitute. Block centroids were set as incidents (demand locations) and facilities were set as facilities (supply locations). The number of facilities to find per incident was set to five, ensuring that each block centroid was connected to up to five facilities within the 10-minute impedance cutoff. Setting this value above one is critical: using the default of one would return only the nearest facility, which would underestimate accessibility for locations near multiple facilities and would not replicate the logic of the 2SFCA's second step. This approach is conceptually consistent with the non-boundary implementation described by Bryant and Delamater (2019), in which the floating catchment is operationalized through enumeration of reachable supply-demand pairs from a route table rather than through explicit service area polygon generation.

The resulting Routes table was exported and joined to facility attributes, allowing each route record to carry the R_j value of its associated facility. The table was then summarized by block centroid (IncidentID), summing all R_j values associated with each centroid to produce the final A_i score. These scores were joined back to the block centroid point layer and subsequently to the block polygon layer for cartographic display.

3.3 Catchment Size Justification and MAUP Considerations

A 10-minute driving catchment was selected based on the nature of the service being analyzed. Early childhood care is a daily-use service for working families, which means travel time sensitivity is high, parents dropping children off before work are unlikely to travel large distances. While some studies of healthcare accessibility use 30- or 60-minute catchments (Becker 2016; Luo and Wang 2003), childcare operates on a different logic: it is used daily and must be integrated into a morning commute routine. A 10-minute driving catchment is therefore more defensible for this context than the longer catchments typical of physician-shortage studies.

The Modifiable Areal Unit Problem (MAUP), a well-known challenge in spatial analysis in which the results of an analysis can change depending on how the geographic units are defined and scaled — is present throughout this analysis. Population demand is measured at the census block level, which is the finest available Census geography, minimizing but not eliminating aggregation error. As Bryant and Delamater (2019) demonstrated, using block-level rather than tract-level data substantially reduces the misrepresentation of individual-level accessibility, though some error persists because the centroid of a block may not accurately represent the location of all residents within it. Additionally, the city boundary creates edge effects: facilities just outside San Francisco County (e.g., in Daly City to the south) are excluded from the analysis, which may cause the accessibility scores of blocks near the city's southern edge to be underestimated.

4. Results

The 2SFCA analysis produced accessibility scores (A_i) for 997 census block centroids across San Francisco County. Scores range from 0.00285 to 0.01144, classified into five intervals — Very Low, Low, Moderate, High, and Very High Access — using a natural breaks (Jenks) classification scheme, which groups values by minimizing within-class variance. These results are displayed in Figure 3 (Childcare Accessibility Score — 10-Minute Driving Catchment) and discussed in relation to demand and supply patterns shown in Figures 4 through 7.

4.1 Spatial Pattern of Accessibility Scores

The map of A_i scores (Figure 3) reveals pronounced geographic inequality in childcare accessibility across San Francisco. The highest accessibility scores (Very High Access, $A_i \geq 0.00783$) are concentrated in a central band running roughly from the Western Addition and

Mission districts eastward through South of Market. This corridor coincides with areas that have both moderate-to-high child populations and a relatively dense cluster of early childhood facilities, meaning that families in these neighborhoods have access to multiple facilities without any single facility facing overwhelming competition from the surrounding population.

By contrast, the lowest accessibility scores (Very Low Access, $A_i \leq 0.00285$) are concentrated in two distinct zones: the western districts, particularly the Sunset, Richmond, and the area surrounding Golden Gate Park and the Presidio, and portions of the southeastern neighborhoods, including parts of Bayview-Hunters Point. These areas share a common characteristic: either they have relatively few facilities within a 10-minute drive of their block centroids, or the facilities that are reachable serve very large surrounding child populations, producing low R_j values that translate into low cumulative A_i scores.

Childcare Accessibility Score (A_i) — 10-Minute Driving Catchment

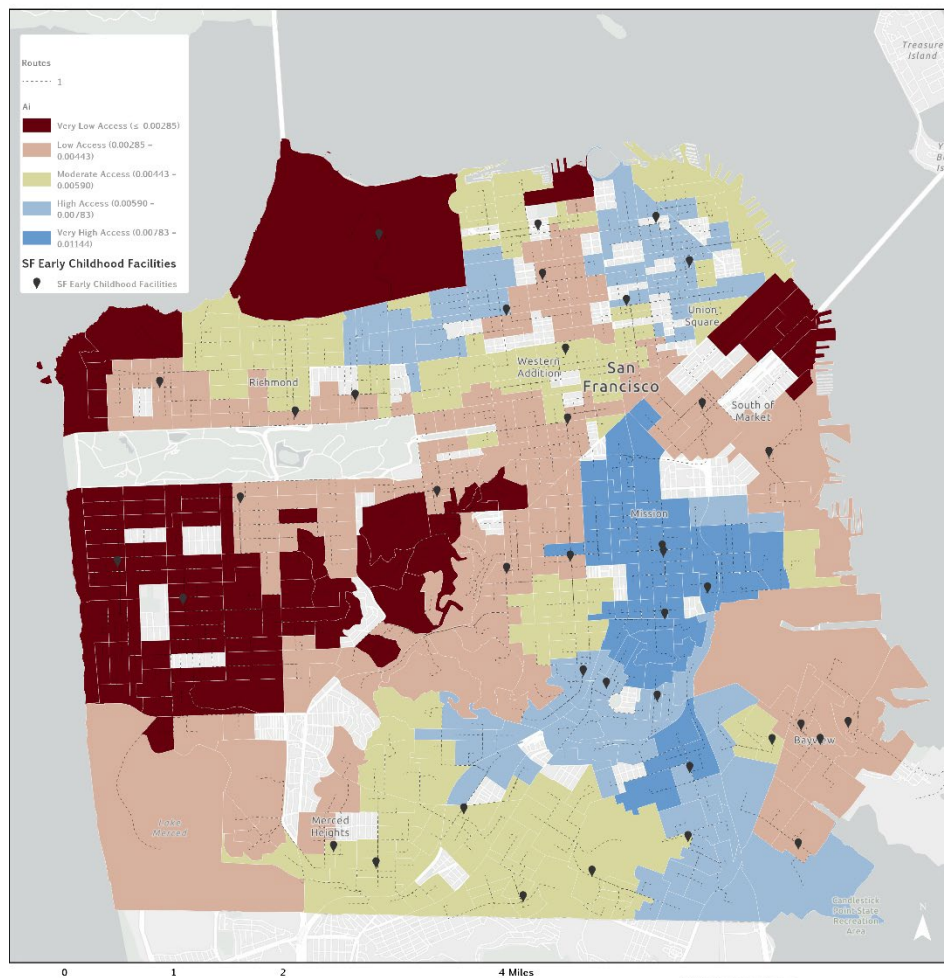


Figure 3. Childcare Accessibility Score (A_i) — 10-Minute Driving Catchment. Five accessibility classes displayed by census block, San Francisco County. Higher scores (blue) indicate greater access; lower scores (dark red) indicate underserved areas. Source: Author's analysis.

4.2 Supply-to-Demand Ratios (R_j) by Facility

Figure 4 (Childcare Facility Capacity: Supply-to-Demand Ratio) displays the R_j values calculated for each facility in Step 1 of the 2SFCA. A facility with a high R_j value has relatively few children within its 10-minute catchment, meaning that its capacity is distributed across a smaller competing population. A facility with a low R_j value is effectively more burdened — surrounded by a larger child population relative to what it can supply.

The map shows that facilities in the western part of the city — in the Richmond and Sunset districts — tend to have higher R_j values (shown as large blue circles), indicating they serve comparatively smaller child populations relative to their supply. This might initially seem to suggest strong access, but the accessibility score map (Figure 3) tells a different story: these western neighborhoods still score low on A_i because very few facilities fall within a 10-minute drive of most block centroids there. There are fewer facilities total, even if individual ones face less demand pressure. Meanwhile, facilities in the Mission, Bayview, and South of Market areas tend to have lower R_j values, reflecting the high density of young children in those neighborhoods. This pattern directly drives the low A_i scores visible in Figure 3 for Bayview-Hunters Point: although there are some facilities in the area, the ones present are overwhelmed by the surrounding child population.

Childcare Facility Capacity (Rj: Supply-to-Demand Ratio)

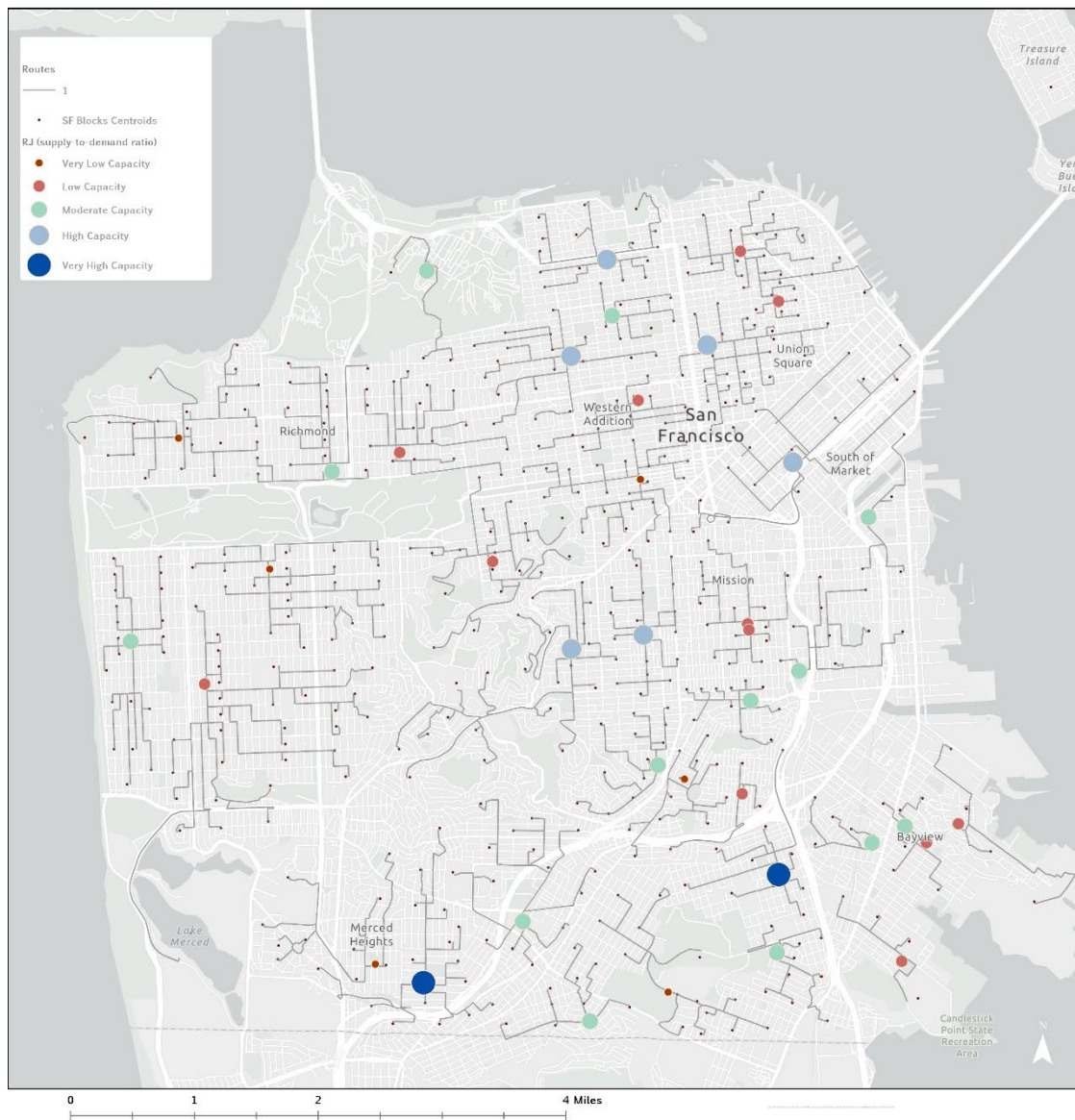


Figure 4. Childcare Facility Capacity (Rj: Supply-to-Demand Ratio). Each dot represents a licensed early childhood facility; dot size and color indicate the supply-to-demand ratio computed in Step 1 of the 2SFCA. Larger, darker blue dots indicate higher capacity relative to surrounding demand. Source: Author's analysis.

4.3 Demand and Network Structure

Figure 5 (Childcare Demand per Facility: Ages 0–5) illustrates the total population of children ages zero to five captured within each facility's 10-minute catchment. Facilities in Bayview-Hunters Point and the Mission district serve the largest child populations (circles sized 1,607–2,035 children), while facilities in the western and northern neighborhoods serve

substantially fewer. This demand concentration in the southeast is a key driver of the low A_i scores visible there.

Figure 6 (Childcare Accessibility Network: 10-Minute Driving Time) shows the route structure produced by the Closest Facility solver. Routes are classified by travel time band: those requiring fewer than four minutes are the thickest and highest-confidence connections, while routes of six to ten minutes are thinner. The network visualization reveals that block centroids in the Sunset and Richmond districts are either generating very few routes or routes that extend to more distant facilities, consistent with the low A_i scores observed in those areas. The Bayview neighborhood shows dense route connections but to facilities with low R_j values, confirming that the problem there is not physical distance but rather facility capacity relative to demand.

Childcare Demand per Facility (Ages 0-5)

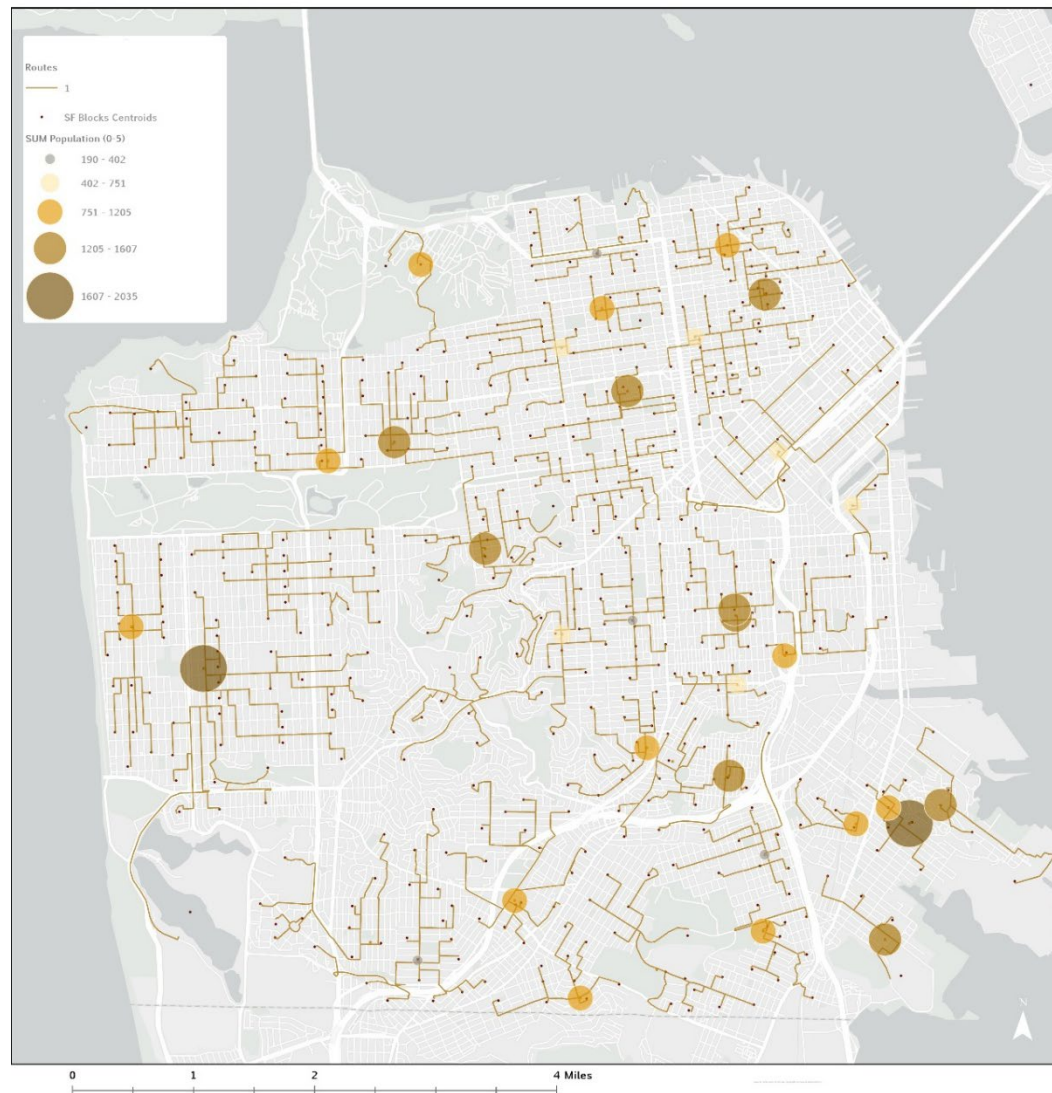


Figure 5. Childcare Demand per Facility (Ages 0–5). Proportional circles at each facility location show the total population of children 0–5 within its 10-minute driving catchment. Source: Author's analysis.

Childcare Accessibility Network (10-minute Driving Time)

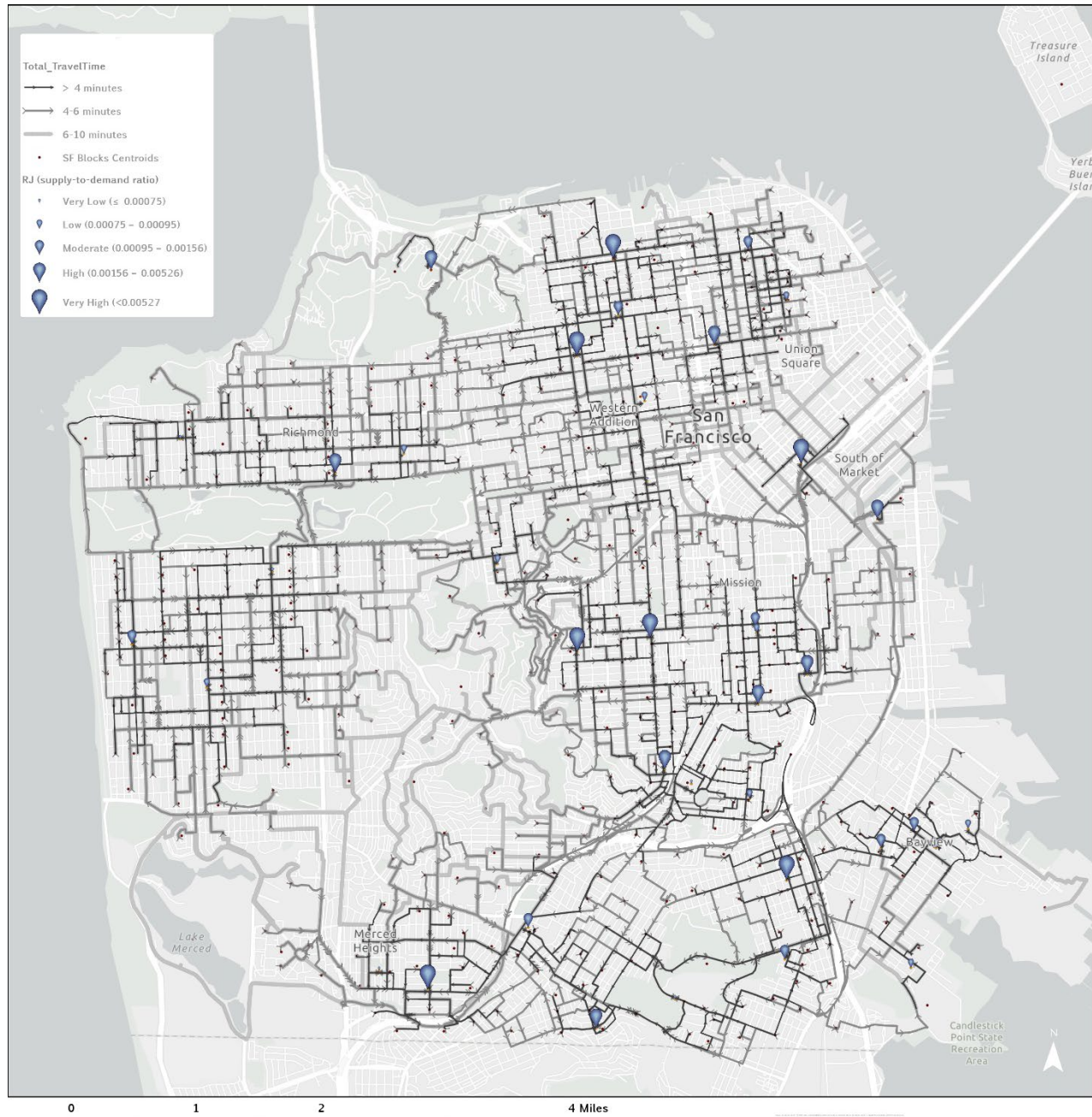


Figure 6. Childcare Accessibility Network (10-Minute Driving Time). Routes from block centroids to reachable facilities, classified by travel time. Facility markers sized by supply-to-demand ratio (Rj). Source: Author's analysis.

4.4 Accessibility vs. Demand Comparison

Figure 7 (Accessibility vs. Demand) overlays the Ai accessibility scores with the distribution of children ages zero to five using a bivariate color scheme — pink tones represent child population density and blue tones represent the Ai score. Blocks that appear deep blue-purple are both densely populated and well-served; blocks that appear pink or light-colored are either densely populated but poorly served, or sparsely populated. The map highlights a cluster of light pink and lavender blocks in Bayview-Hunters Point and parts of the Mission — areas with high child populations but comparatively weaker accessibility scores — suggesting these are the most critically underserved zones in the city from a childcare access standpoint.

Accessibility vs Demand

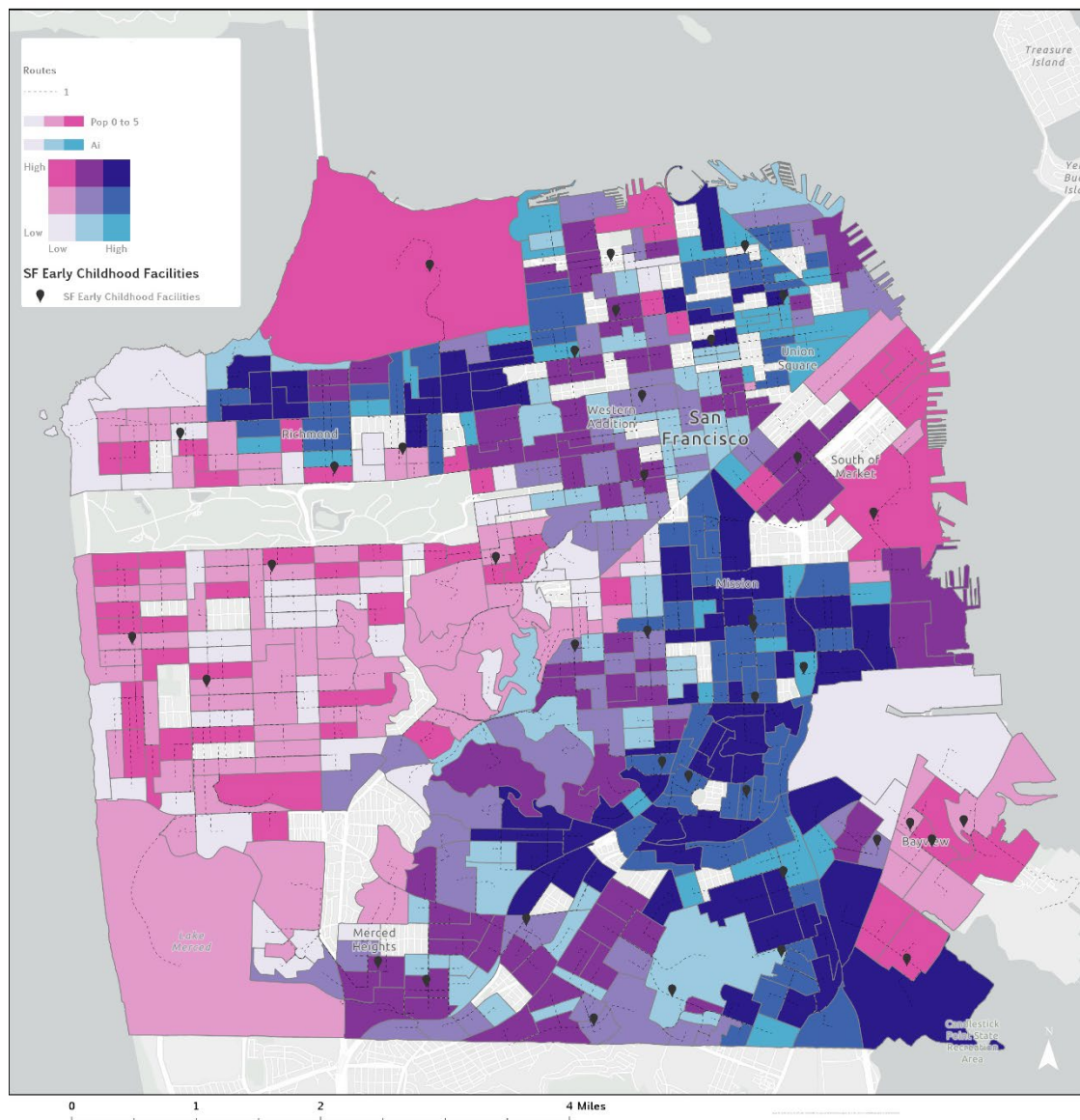


Figure 7. Accessibility vs. Demand (Bivariate Map). Pink tones indicate higher population of children 0–5; blue tones indicate higher accessibility score (A_i). Areas appearing predominantly pink with low A_i values represent the most underserved neighborhoods. Source: Author's analysis.

Facility Catchment

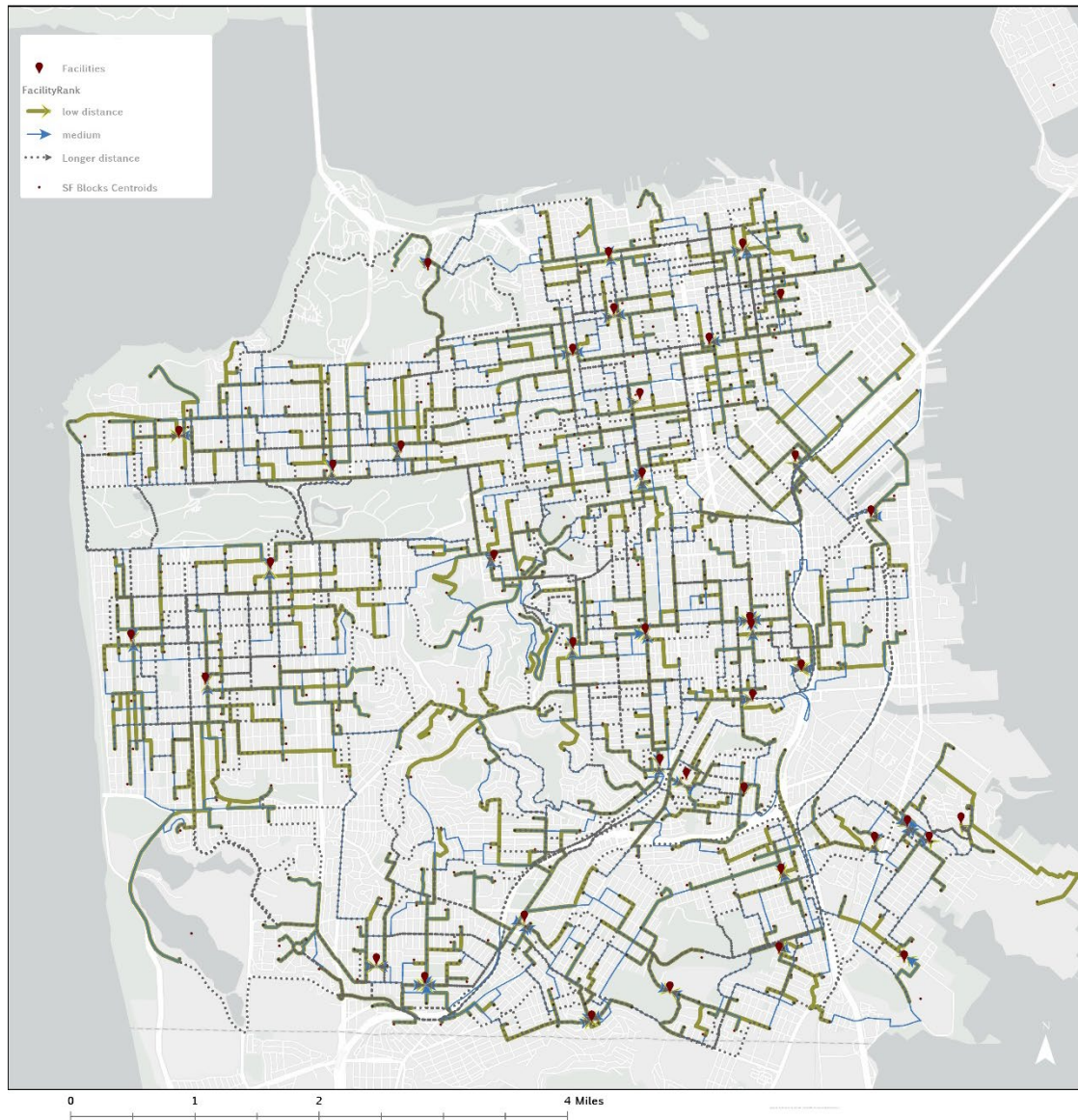


Figure 8. Facility Catchment — Routes from Block Centroids to Facilities by Distance Rank. Green arrows indicate low-distance connections; blue arrows indicate medium-distance; dashed lines indicate longer-distance connections near the 10-minute threshold. Source: Author's analysis.

5. Discussion

5.1 *Real-World Meaning of Results*

The results of this analysis carry concrete implications for families in San Francisco. Neighborhoods such as Bayview-Hunters Point, parts of the Mission, and the western Sunset and Richmond districts emerge as the most underserved areas for early childhood care access. These are not random findings: Bayview-Hunters Point is one of San Francisco's historically lower-income communities of color, with a younger population profile and fewer community services relative to wealthier neighborhoods. The finding that this area faces both high demand and low accessibility scores aligns with broader patterns of urban inequality in which residents with the greatest need often have the least access to services.

The western neighborhoods, Sunset and Richmond, present a different type of underservice. These areas have comparatively fewer young children, and the 10-minute driving catchment may be capturing a real behavioral constraint: in a dense urban environment with significant traffic, a 10-minute drive covers a smaller geographic area than it would in a suburban or rural setting. Families in these neighborhoods who rely on car travel may face reasonable commuting options, but those without vehicles face a more severe access barrier that this car-based analysis does not fully capture.

The findings echo those of related studies. Luo and Wang (2003) demonstrated that spatial accessibility analyses consistently reveal local pockets of poor access that aggregate regional measures obscure. In the San Francisco childcare context, a city-wide statistic — such as the number of licensed childcare slots per 100 children — would mask the extreme heterogeneity in access quality across neighborhoods that the 2SFCA reveals. Similarly, the pattern observed here, where facility-dense areas still underserve their populations due to high competing demand, mirrors findings from health care accessibility studies in which the presence of facilities does not guarantee adequate per-capita access (Luo and Qi 2009).

5.2 *MAUP and Analytical Limitations*

Several limitations of this analysis deserve candid discussion. First, the MAUP operates at the scale of the census block. Using blocks rather than block groups reduces but does not eliminate aggregation error. The centroid of a block represents the average location of all residents within it, which introduces positional inaccuracy, particularly for large, irregularly shaped blocks common

in the Sunset and Richmond districts. As Bryant and Delamater (2019) showed, block-level aggregation produces FCA measurements more accurate than block-group or tract-level data, but individual-level data would be needed to fully eliminate this source of error.

Second, the 10-minute catchment threshold is a blunt instrument. The standard 2SFCA treats all residents within 10 minutes as equally accessible to a facility, and all residents beyond 10 minutes as completely inaccessible, a dichotomous assumption that the Enhanced 2SFCA (E2SFCA) addresses by applying distance-decay weights within the catchment. Luo and Qi (2009) showed that applying Gaussian distance-decay weights (e.g., 1.0 for 0–10 minutes, 0.68 for 10–20, and 0.22 for 20–30 minutes) produces accessibility patterns more consistent with real-world behavior, because people closer to a facility are more likely to use it. A future improvement to this analysis would apply the E2SFCA framework, which would likely identify higher access scores in the immediate vicinity of facilities and lower scores at the edge of catchments.

Third, edge effects introduce systematic underestimation of access for blocks near the city's southern and western boundaries. Facilities in neighboring Daly City and the northern San Mateo County communities, which may be equally or more accessible by car than some San Francisco facilities — are excluded because the analysis is bounded by county lines. This is a common limitation of 2SFCA studies applied to a single administrative jurisdiction, and it is particularly consequential here because San Francisco's southern neighborhoods (Excelsior, Visitacion Valley) are geographically close to Daly City.

Fourth, the supply variable used here, facility count, with each facility weighted equally, does not account for differences in enrollment capacity, operational hours, cost, subsidy acceptance, or language of instruction. A facility licensed for 12 children operates very differently from one licensed for 80 children, yet both contribute equally to the S_j term in this analysis. Incorporating licensed capacity data, which is available in the full DataSF dataset, would substantially improve the accuracy of the R_j calculation and is the most impactful single improvement that future analysis should prioritize.

Finally, the workaround of using the Closest Facility solver in place of the OD Cost Matrix, while methodologically sound, introduces a practical limitation: by setting the number of facilities to find to five, the analysis may miss connections to a sixth, seventh, or more facilities within the catchment for blocks with many nearby options. In a facility-dense area like the Mission district,

this could slightly underestimate A_i scores. Setting the number higher (e.g., 10) would reduce this risk but would increase processing time and credit usage.

5.3 Assessment of the 2SFCA Method

The 2SFCA method provides a meaningful and practical improvement over simpler accessibility measures. Unlike proximity-only measures, which identify simply how far the nearest facility is, the 2SFCA incorporates competition: it recognizes that a facility surrounded by thousands of children in need is functionally less accessible than a facility of the same size surrounded by hundreds. This is a theoretically important advance, because it connects the concepts of supply and demand in a spatially explicit way that mirrors how resource scarcity actually operates in a city.

At the same time, the method has genuine blind spots. It does not capture non-spatial barriers to access, including cost, subsidy availability, linguistic compatibility, cultural acceptability, or parental work schedules, factors that Luo and Wang (2003) and subsequent scholars have acknowledged as essential components of realized accessibility. A family that lives within 10 minutes of a facility they cannot afford is not, in any meaningful sense, well-served by that facility. The 2SFCA produces a measure of potential spatial accessibility, not actual utilization. This distinction is essential for policymakers: high A_i scores do not guarantee high enrollment rates, and low A_i scores, while a red flag, may reflect non-spatial barriers rather than geographic ones alone.

In sum, the 2SFCA is a powerful tool for identifying which parts of a city are spatially underserved relative to their population needs. When applied to childcare access in San Francisco, it reveals a city in which geographic access to early childhood services is profoundly unequal, with the most socioeconomically vulnerable neighborhoods facing the most severe supply-demand imbalances. These findings can usefully inform city planning decisions about where to prioritize new facility investments or childcare subsidy expansion, while the method's limitations remind analysts that spatial access is a necessary but not sufficient condition for equitable service delivery.

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Appendix: Workflow Diagram

The following diagram summarizes the step-by-step workflow implemented in ArcGIS Pro to execute the 2SFCA analysis.

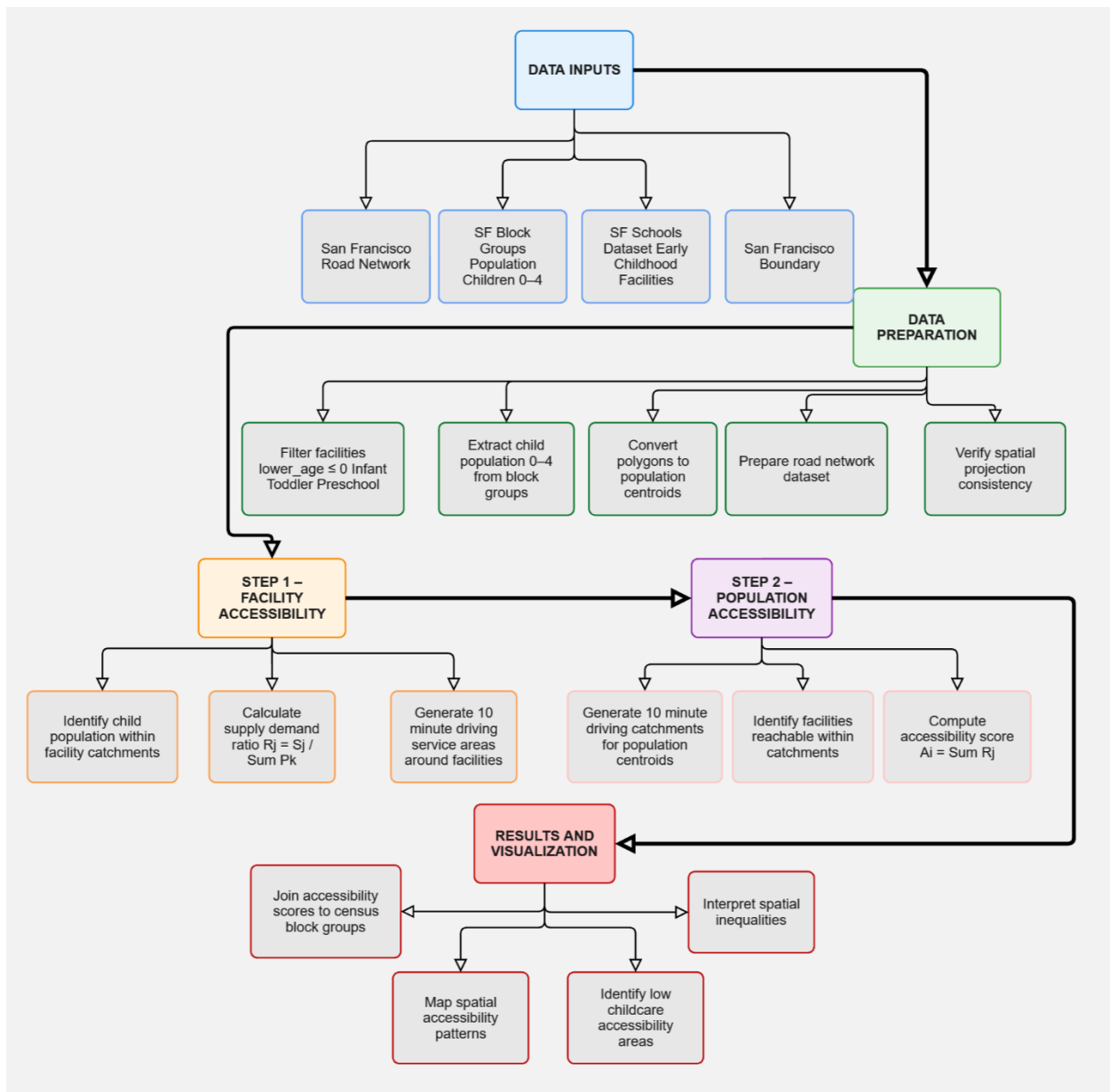


Figure A1. Analytical Workflow Diagram — 2SFCA Implementation Using ArcGIS Pro Closest Facility Solver. Steps 1–2 correspond to the two-step floating catchment area procedure; intermediate steps detail data preparation, network solving, R_j calculation, and A_i aggregation.